

# Recipe Recommender — Data Infrastructure & Access Guide

Data Ingestion, Validation, Splitting & Versioning (Assignment Steps 1–4)

Prepared by Amir Farooq · August 4, 2026 · ML Ops Final Project — Recipe Recommendation System

---

## Overview

This covers the data infrastructure for our recipe recommender MLOps project — specifically my four rows on the assignment sheet: the dataset download script, schema validation, the train/validation/test split, and DVC data versioning. Preprocessing, feature engineering, model training, and everything downstream builds on the versioned data this produces, so this doc explains what's already set up, how to get access to it, and what to expect once the scripts are built.

## Infrastructure Already Set Up

| Resource        | Value                                                               |
|-----------------|---------------------------------------------------------------------|
| GCP Project ID  | recipe-mlops                                                        |
| Storage Bucket  | recipe-mlops-dvc (multi-region US, private — public access blocked) |
| Service Account | dvc-storage-access@recipe-mlops.iam.gserviceaccount.com             |
| Raw Data Source | Kaggle — thedevastator/better-recipes-for-a-better-life             |
| Versioning Tool | DVC (Data Version Control), remote pointed at the bucket above      |

## What I'm Building (Assignment Steps 1–4)

| Step | Task                                    | Deliverable                                    |
|------|-----------------------------------------|------------------------------------------------|
| 1    | Build dataset download script           | data/ingest.py + raw dataset files             |
| 2    | Create schema validation                | data/validate.py + validation report           |
| 3    | Create train / validation / test splits | Versioned train, validation, and test files    |
| 4    | Configure DVC                           | dvc.yaml, .dvc files, and remote configuration |

## How the Pipeline Flows

Kaggle dataset → **ingest.py** downloads the raw recipe data → **validate.py** checks schema and data quality → split script produces isolated train / validation / test sets → DVC tracks and versions every stage → the versioned data is pushed to the **recipe-mlops-dvc** bucket. Everyone else pulls that same versioned copy instead of re-downloading or re-processing anything themselves — that's what keeps the whole team working from identical data.

## Getting Access (For Teammates)

### What I'll do (once, per teammate):

- Add your Google account under IAM & Admin → IAM on the recipe-mlops GCP project, with the Storage Object Viewer or Storage Admin role. Send me the Google account email you want added.

### What you'll need to do (one-time setup on your end):

- 1 Send Amir the Google account email you want added to the project.
- 2 Install the Google Cloud CLI: `cloud.google.com/sdk/docs/install`
- 3 Run `gcloud auth application-default login` and sign in with the account Amir added — this lets DVC authenticate on your machine without needing a separate key file.
- 4 Run `gcloud config set project recipe-mlops`
- 5 Install DVC with the GCS extension: `pip install "dvc[gs]"`
- 6 Once the project files are shared (Drive for now, GitHub repo later), run `dvc pull` from inside the project folder — this downloads the exact versioned data my pipeline produced.

*Note: you do **not** need your own Kaggle account or API token. Only the ingestion script (which only I run) talks to Kaggle directly — everyone else works from the versioned copy in the GCS bucket.*

## What You'll Find in the Project (Once Built)

Planned structure — file names may shift slightly as it comes together:

```
project-root/
  data/
    ingest.py      # downloads raw data from Kaggle
    validate.py    # schema / data-quality checks
    raw/           # DVC-tracked raw files
    splits/        # train.csv, val.csv, test.csv (DVC-tracked)
  dvc.yaml         # pipeline stage definitions
  .dvc/config      # remote (GCS bucket) pointer
  requirements.txt
```

## Status & Next Steps

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| Done        | GCP project, bucket, and service account created                  |
| Done        | Kaggle dataset identified and access confirmed                    |
| In progress | Ingestion, validation, and split scripts                          |
| Pending     | DVC pipeline wired up and first data version pushed to the bucket |
| Pending     | Teammates added to GCP IAM (send me your Google account email)    |

*I'll share an updated version of this doc with exact commands and confirmed file locations once the scripts are built and the first data version is pushed to the bucket. Questions in the meantime — just ask.*